Mobility and Automotive Tech

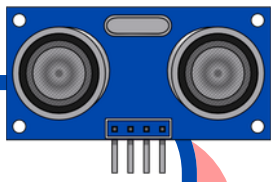


Project Name: Emergency Braking System



The Problem: Rohan is testing a small robot car on a road model. 🚗 Suddenly, a box appears in front of the car, but the car is moving too fast to stop by itself. Rohan thinks the car should be able to detect an obstacle before it gets too close. He decides to build a smart system that can sense the obstacle and stop the car automatically.

🎯 **What You Will Build:** Students will build a smart emergency braking system that continuously checks the road ahead using an ultrasonic sensor. When an obstacle is detected at a dangerous distance, the system will stop the vehicle and alert the driver.



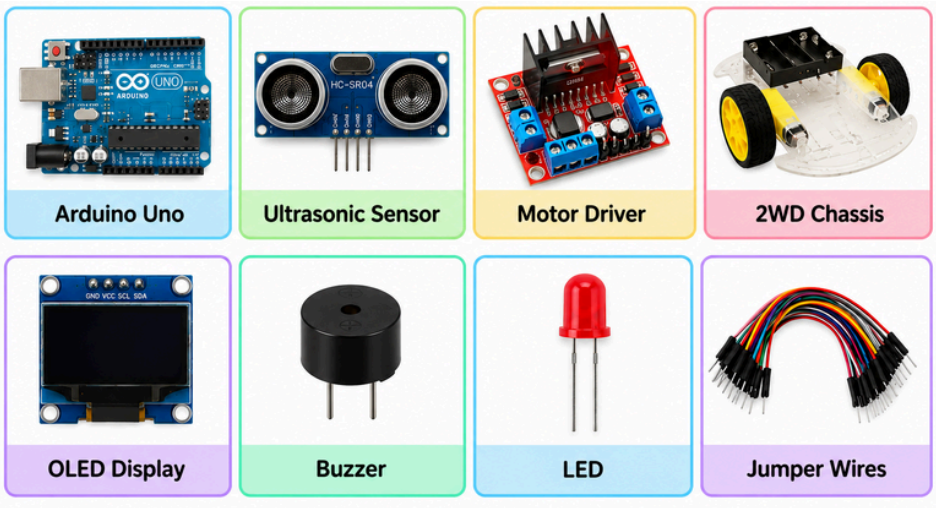


📡 What is an ultrasonic sensor?

An ultrasonic sensor measures distance by sending sound waves and receiving their echoes back. It helps detect how far an object is from the sensor.

Example: 🦇 Just like a bat uses sound waves to find objects while flying in the dark, an Ultrasonic Sensor uses sound waves to detect obstacles.

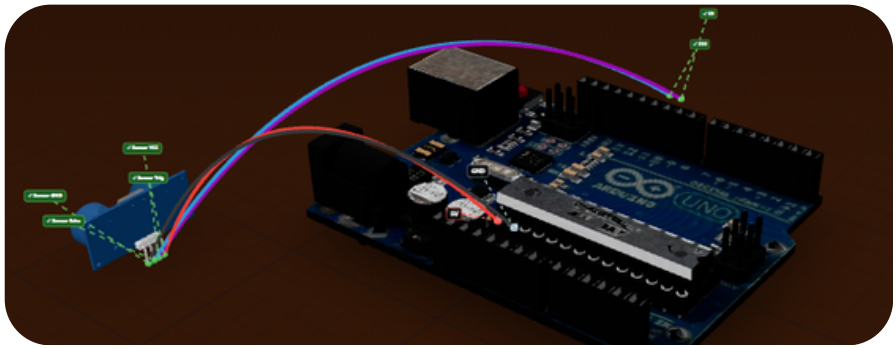
Components Required



Circuit Designing

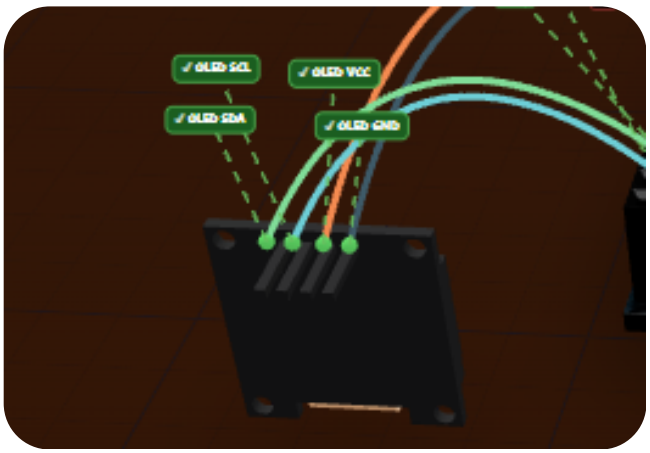
Step 1: Connect the ultrasonic sensor to the Arduino Uno.

Sensor Pins	Arduino Pins
Vcc Pin	5V Pin
GND Pin	GND Pin
Trig Pin	Digital Pin 9
Echo Pin	Digital Pin 10



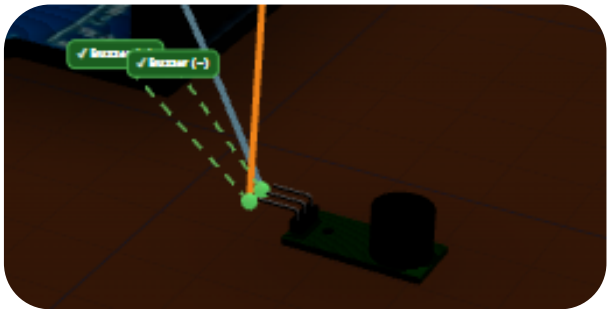
Step 2: Connect the OLED display to the Arduino Uno.

OLED Pins	Arduino Pins
Vcc Pin	5V Pin
GND Pin	GND Pin
SDA Pin	Analog A4 Pin
SCL Pin	Analog A5 Pin



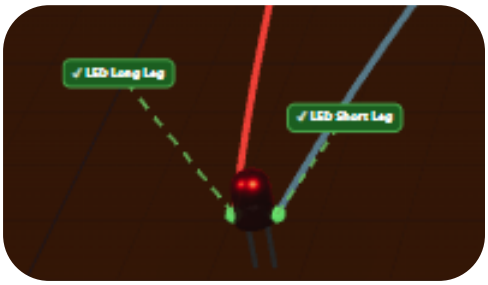
Step 3: Connect the buzzer to the Arduino Uno.

Buzzer Pins	Arduino Pins
Positive Pin	Digital Pin 7
Negative Pin	GND Pin



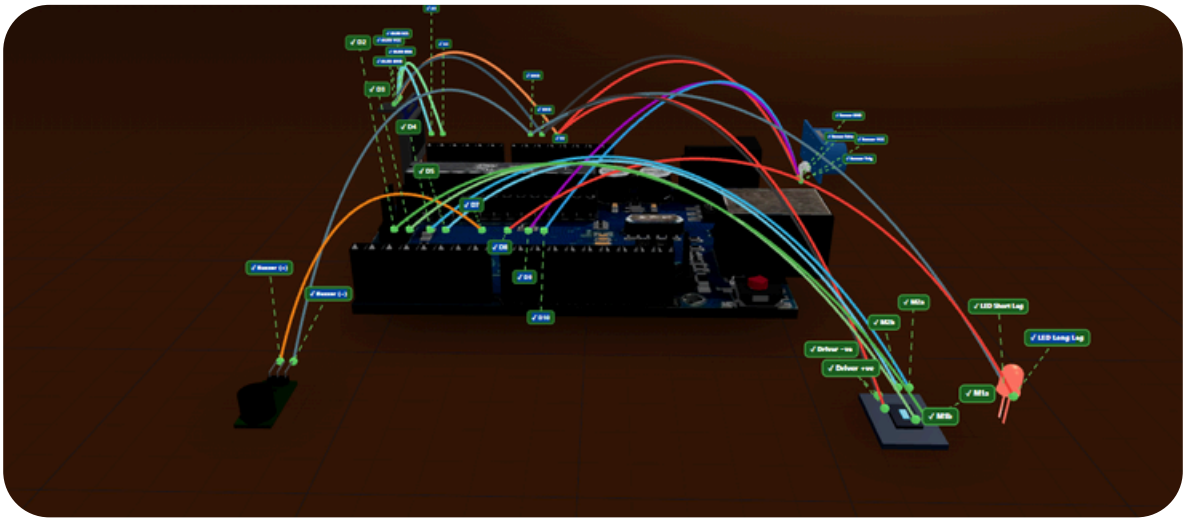
Step 4: Connect the LED to the Arduino Uno.

LED Pins	Arduino Pins
Long Leg	Digital Pin 8
Short Leg	GND Pin



5. Connect the motor driver module to the Arduino Uno.

Motor Driver Pins	Arduino Pins
M1a Pin	Digital Pin 2
M1b Pin	Digital Pin 3
M2a Pin	Digital Pin 4
M2b Pin	Digital Pin 5
+ve Pin	5V Pin
-ve Pin	GND Pin



Project Coding

1. Open the Arduino Uno software and write code to add the OLED libraries.

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

#define TRIG_PIN 9
#define ECHO_PIN 10
#define MOTOR_PIN1 2
#define MOTOR_PIN2 3
#define MOTOR_ENABLE 4
#define BUZZER_PIN 7
#define LED_PIN 8
#define OLED_RESET -1
Adafruit_SSD1306 display(OLED_RESET);

const int dangerDistance = 20; // Distance in cm
```

2. Make code to define the input and the output pins.

```
void setup() {
  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);
  pinMode(MOTOR_PIN1, OUTPUT);
  pinMode(MOTOR_PIN2, OUTPUT);
  pinMode(MOTOR_ENABLE, OUTPUT);
  pinMode(BUZZER_PIN, OUTPUT);
  pinMode(LED_PIN, OUTPUT);

  Serial.begin(9600);
  display.begin(SSD1306_SWITCHCAPVCC, 0x3C);
  display.clearDisplay();
  display.setTextSize(1);
  display.setTextColor(WHITE);
```

3. Make the code to start the motor.

```
  // Start the motor
  digitalWrite(MOTOR_PIN1, HIGH);
  digitalWrite(MOTOR_PIN2, LOW);
  digitalWrite(MOTOR_ENABLE, HIGH);
}
```

4. Make code to calculate the sensor distance and print the distance on the OLED display.

```
47  void loop() {
48    long duration, distance;
49    digitalWrite(TRIG_PIN, LOW);
50    delayMicroseconds(2);
51    digitalWrite(TRIG_PIN, HIGH);
52    delayMicroseconds(10);
53    digitalWrite(TRIG_PIN, LOW);
54    duration = pulseIn(ECHO_PIN, HIGH);
55    distance = (duration / 2) / 29.1;
56    display.clearDisplay();
57    display.setCursor(0, 0);
58    display.print("Distance: ");
59    display.print(distance);
60    display.println(" cm");
```

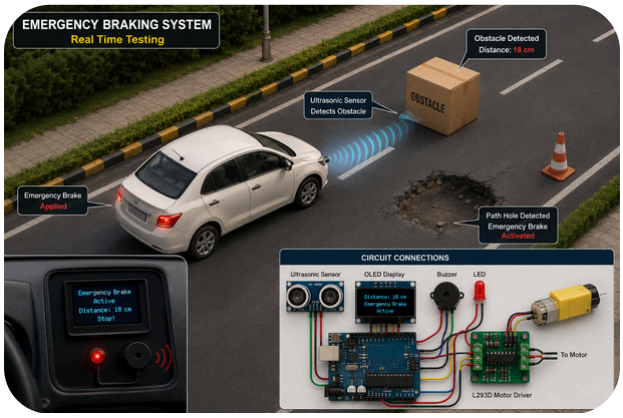
5. Make code to use the conditions to check the obstacles in front of the sensor and display the obstacle message on the OLED display.

```
62    if (distance < dangerDistance) {
63      display.println("Obstacle Ahead!");
64      display.println("Emergency Brake");
65      digitalWrite(BUZZER_PIN, HIGH);
66      digitalWrite(LED_PIN, HIGH);
67      digitalWrite(MOTOR_ENABLE, LOW); // Stop motor
68    } else {
69      display.println("Status: Safe");
70      digitalWrite(BUZZER_PIN, LOW);
71      digitalWrite(LED_PIN, LOW);
72      digitalWrite(MOTOR_ENABLE, HIGH); // Run motor
73    }
```

6. Make code to display all the sensor values on the OLED display.

```
74
75    display.display();
76    delay(100);
77  }
```

Proejct Output





🌾 Where Is This Used in Real Life?

Similar emergency braking systems are used in modern electric cars. vehicles, and advanced driver assistance systems (ADAS). These systems help drivers avoid collisions by detecting obstacles

Learning Outcomes

- The students will be able to measure obstacle distance using an Ultrasonic Sensor.
- The students will be able to code to control a DC motor using an L293D Motor Driver.
- The students will be able to create an automatic emergency braking system using sensors and coding.



Section 1: Multiple Choice Questions

1. What problem does Rohan want to solve?

a) The car cannot change its colour
b) The car may not stop when an obstacle is nearby
c) The car needs more wheels
d) The car needs a bigger battery

2. Which sensor helps the car detect an obstacle ahead?

a) Soil Sensor
b) Temperature Sensor
c) Ultrasonic Sensor
d) Touch Sensor

3. What happens when an obstacle is too close?

a) The motor runs faster
b) The emergency brake stops the motor
c) The sensor turns off
d) The car changes colour

Section 2: True and False Statements

1. An ultrasonic sensor can help detect how far an obstacle is.....

2. The system keeps the motor running when an obstacle is at a dangerous distance.....

3. The LED and buzzer can alert the driver when there is danger.....